problem statement

Proposition 0.1. *Define $g_{n,i}(x) = I_x(i, n-i+1)$, we want to prove $g^{-1}_{n,n-j}(x)^n$ is an increasing function with respect to n, or $g_{n,n-j}(x^{\frac{1}{n}})$ is a decreasing function with respect to n. We have known $g_{n,n-j}(x)$ is a decreasing function with respect to n and $g^{-1}_{n,n-j}(x)$ is an increasing function with respect to n.*

I'm going to drop the $g_{i,n}$ notation here because it is very confusing. That notation should only be used when there are space constraints.

We will show that

$$I_{x^{\frac{1}{n}}}(n-j, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, n-j) > 0.$$

Assume $x^{\frac{1}{n}} > j/n$:

Integration by parts allows us to write

$$\begin{aligned}
&I_{x^{\frac{1}{n}}}(n-j, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, n-j) \\
&\quad = I_{x^{\frac{1}{n}}}(n-j+1, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, n-j) + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j, j+1)} \\
&\quad = -\int_{x^{\frac{1}{n}}}^{x^{\frac{1}{n+1}}} \frac{y^{n-j}(1-y)^j}{B(n-j+1, j+1)} dy + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j, j+1)}.
\end{aligned}$$

The function $y^a(1-y)^b$ achieves its maximum value at $y^* = a/(a+b)$. The assumption that $x^{\frac{1}{n}} > j/n$ implies that the function $y^{n-j}(1-y)^j$ is decreasing over $(x^{\frac{1}{n}}, x^{\frac{1}{n+1}})$. This implies that

$$\begin{aligned}
&-\int_{x^{\frac{1}{n}}}^{x^{\frac{1}{n+1}}} \frac{y^{n-j}(1-y)^j}{B(n-j+1, j+1)} dy + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j, j+1)} \\
&\qquad \geq \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}(x^{\frac{1}{n}} - x^{\frac{1}{n+1}})}{B(n-j+1, j+1)} + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j, j+1)}.
\end{aligned}$$

And $B(n-j+1,j+1) = B(n-j,j+1)\frac{j+1}{n+1}$ allows us to express the right hand side of the above inequality as

$$\begin{aligned}
&\frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}(x^{\frac{1}{n}}-x^{\frac{1}{n+1}})}{B(n-j+1,j+1)} + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j,j+1)} \\
&= \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}(x^{\frac{1}{n}}-x^{\frac{1}{n+1}})}{B(n-j,j+1)\frac{j+1}{n+1}} + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(n-j)B(n-j,j+1)} \\
&= \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}}{B(n-j,j+1)}\left(\frac{n+1}{j+1}\left(x^{\frac{1}{n}}-x^{\frac{1}{n+1}}\right)+\frac{1}{n-j}\left(1-x^{\frac{1}{n}}\right)\right) \\
&= \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}}{(n-j)B(n-j,j+1)}\left(\frac{(n+1)(n-j)}{j+1}\left(x^{\frac{1}{n}}-x^{\frac{1}{n+1}}\right)+1-x^{\frac{1}{n}}\right)
\end{aligned}$$

Let

$$f_{n,j}(x) = \frac{(n+1)(n-j)}{j+1}\left(x^{\frac{1}{n}}-x^{\frac{1}{n+1}}\right)+1-x^{\frac{1}{n}}. \tag{1}$$

Now,

$$\begin{aligned}
f'_{n,j}(x) &= \frac{(n+1)(n-j)}{j+1}\left(\frac{x^{\frac{1-n}{n}}}{n}-\frac{x^{-\frac{n}{n+1}}}{n+1}\right)-\frac{x^{\frac{1-n}{n}}}{n} \\
&= \frac{x^{\frac{1-n}{n}}}{n}\left(\frac{(n+1)(n-j)}{j+1}\left(1-\frac{n}{n+1}x^{-\frac{n}{n+1}-\frac{1-n}{n}}\right)-1\right) \\
&= \frac{x^{\frac{1-n}{n}}}{n}\left(\frac{(n+1)(n-j)}{j+1}\left(1-\frac{n}{n+1}x^{-\frac{1}{n(n+1)}}\right)-1\right).
\end{aligned}$$

From here we see that there exists a single root,

$$\begin{aligned}
&\frac{x^{\frac{1-n}{n}}}{n}\left(\frac{(n+1)(n-j)}{j+1}\left(1-\frac{n}{n+1}x^{-\frac{1}{n(n+1)}}\right)-1\right) \overset{\text{set}}{=} 0, \\
&\implies \frac{(n+1)(n-j)}{j+1}\left(1-\frac{n}{n+1}x^{-\frac{1}{n(n+1)}}\right)=1, \\
&\implies \frac{n}{n+1}x^{-\frac{1}{n(n+1)}} = 1-\frac{j+1}{(n+1)(n-j)}, \\
&\implies x^* = \left(\frac{n+1}{n}\left(1-\frac{j+1}{(n+1)(n-j)}\right)\right)^{-n(n+1)}.
\end{aligned}$$

Simplifying further yields

$$\begin{aligned}
\left(\frac{n+1}{n}\left(1-\frac{j+1}{(n+1)(n-j)}\right)\right)^{-n(n+1)} &= \left(\frac{n+1}{n}-\frac{j+1}{n(n-j)}\right)^{-n(n+1)} \\
&= \left(\frac{(n+1)(n-j)}{n(n-j)}-\frac{j+1}{n(n-j)}\right)^{-n(n+1)}
\end{aligned}$$

$$= \left(\frac{n(n-j)+n-2j+1}{n(n-j)} \right)^{-n(n+1)}.$$

We have $x^* \geq 1$ whenever $n-2j+1 \leq 0$. We know that $f_{n,j}(0) = 1$ and $f_{n,j}(1) = 0$, and this implies that $f(x) > 0$ for all $x \in (0,1)$. Now consider when $n-2j+1 > 0$. We can rewrite (1) as

$$f_{n,j}(x) = \frac{(n+1)(n-1)-(j+1)}{j+1} x^{\frac{1}{n}} - \frac{(n+1)(n-1)}{j+1} x^{\frac{1}{n+1}} + 1.$$

From here we can see that $f_{n,j}(x)$ is monotonic by noting that $n-2j+1 = (n-j)-(j+1)$. Therefore, when $x^{\frac{1}{n}} > j/n$ we can conclude that

$$I_{x^{\frac{1}{n}}}(n-j, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, n-j) \geq 0.$$

Assume $x^{\frac{1}{n+1}} < j/n$: